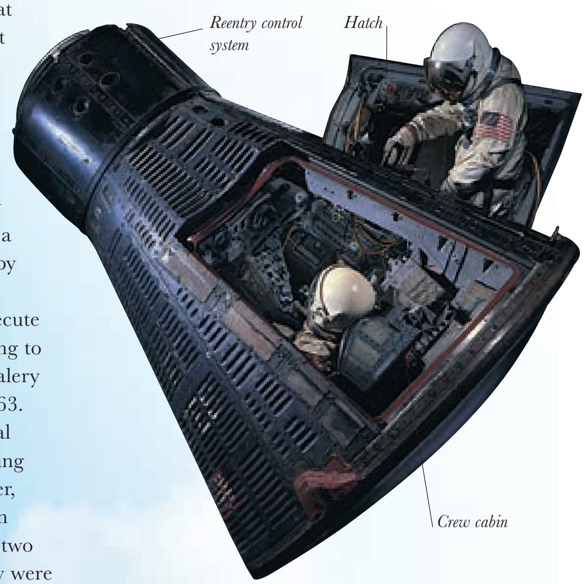
another human cannonball loop through space, but a genuine orbital space flight. Glenn went around the Earth three times in a flight that lasted 4 hours and 55 minutes. It was far from being a sightseeing trip. There were unnerving problems with some automatic systems and at one point ground control began to think that the capsule's heat shield might have become loose. As Friendship 7 reentered the atmosphere, the whole of America held its breath. Even Glenn went through a moment when he believed his vehicle was burning up. But splashdown was successfully accomplished and the astronaut returned to a tickertape parade down Broadway believed by some to be the biggest since Lindbergh.

The Soviets continued to execute the longest space flights, extending to an hour short of five days for Valery Bykovsky in Vostok 5 in June 1963. They also went on scoring political points. While Bykovsky was still circling the Earth, a former cottonmill worker, Valentina Tereshkova, was sent up in Vostok 6. The Soviets not only had two people in orbit at once – a feat they were performing for the second time – but the first woman in space. In this respect, as it turned out, they were two decades ahead of the United States.

The longest flight under the Mercury program was 34 hours 20 minutes, by Gordon Cooper in May 1963. He was the sixth of the Mercury 7 to go into space, and the last under the banner of that program. Only Deke Slayton had missed out, grounded with a minor heart problem. Slayton finally made it into space in 1975.

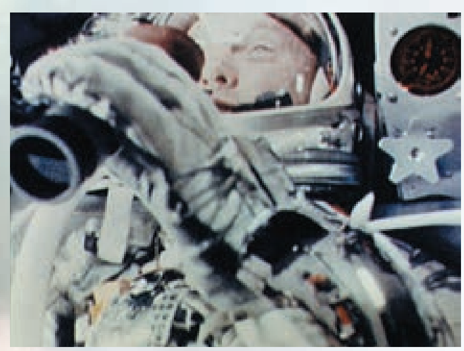
GEMINI 4 SPACECRAFT

The two-man Gemini spacecraft was a major step forward from the Mercury capsules. Weighing 7,000lb (3,200kg), it could be controlled by its pilots to perform elaborate maneuvers in space. Hatches swung open to provide an easy exit for spacewalks.



The announcement of the goal for a manned Moon mission increased the importance of the Mercury project; it was really the first step in learning about human spaceflight. But there was no straight line from the Mercury capsules and their launchers to a Moon voyage. The lunar mission would require not just a far more powerful booster rocket, but also spacecraft with onboard power systems and computers capable of elaborate precision maneuvers such as docking with another craft in space. It would also require astronauts to be trained to spend much longer periods in space and to operate outside their space vehicles – so they could walk on the airless Moon. It was to develop these skills and techniques that NASA embarked on the Gemini project, while longer-term work on the Moon program proper, dubbed Apollo, gathered momentum.

By 1964 the US space program had turned into one of the largest-scale endeavors in the country's history. Yet the Soviets still gave the impression of holding their lead. As NASA worked toward



GLENN IN ORBIT

John Glenn prepares to view the Sun through a photometer during one of three sunsets he witnessed in his five-hour journey through space in February 1962. The astronaut saw what looked like fireflies during his flight, but the "spacebugs" turned out to be particles of ice shaken from the surface of the capsule.